AGI ALPHA FLYWHEEL — When Capability Creates Opportunity

Transformational Intelligence, Experience-Trained Specialists and the Expanding Frontier of Profitable Work

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1 Evidence companion

This index separates executable evidence from economic hypotheses. The constructed origin supports **$22.83 expected operating contribution** before additional retention and research. The unchanged policy declines all **48 fresh cases**; their quote costs reduce the modeled campaign contribution to **–$323.17**. Actual customer collections are **zero**. Complete actual research cost and profit remain unknown. Paths below are relative to `studies/agi-alpha-shared-observation-v1/`.

1.1 What each evidence level establishes

Level	Evidence and boundary
Locally executed	Compiler, independent finite reference, accounting enumeration, archive transfer and recipient process ran. This establishes software behavior on recorded inputs.
Synthetic technical world	State probabilities, observation likelihoods, deadlines, action costs and risk limits were supplied. Calculated outcomes do not establish physical collision probabilities.
Modeled commerce	Participation, acceptance, withdrawal, default, fees, refunds and cash paths are enumerated assumptions. These are neither customers nor receipts.
Sourced context	Primary sources support domain practice, prior art, model capabilities/prices and a wage benchmark. They do not validate this offering's demand.
Not established	Operational Alpha, incremental retained savings, live sensor performance, buyer financing, production admission, authentication through live ENS, or flight authority.

The research team used AI assistance for analysis, implementation, checking and writing. The reference implementation was delegated within the same session under a procedural separation rule; this is not external institutional replication. The three foundational papers informed formulation changes, joint construction of capability and arrangement, and inheritance with fresh mission checks. Their arguments are motivation, not commercial validation. (MONTREAL.AI 2026c, 2026a, 2026b)

1.2 Versions, hashes and chronology

The study parent is commit `47c17f6aca6b1bb8779dd3e47d5ba5f8af3329f4`. Existing research integration was drawn from `231834269599d85e49cfe1ca9917a1a874522d44`. Preserved source manifests bind the lifecycle closure to `2727262caa52766a78916f2ed81f5afaf668edce` and native runtime to `80b345828f34fd8a6e592977dd17ef37141305f4`. New wrappers do not replace the registry.

All following times are UTC on 27 October 2026. Economic assumptions were frozen at **17:00:18**. The first technical freeze at **17:02:49** followed eight development cases. A variable-name-only lint correction was frozen at **17:03:47**, before protected confirmation. The independent reference and 48-case manifest were frozen at **17:03:54** without inspecting compiler, profile or development answers. Confirmation ran **17:04:26–17:04:29**. A later analytical economics correction was recorded at **17:19:15**; it did not alter the primary event tree or technical confirmation.

The authoritative records are `evidence/technical-freeze-v1_0_1.json`, `evaluation/reference-freeze-v1.json`, `evidence/retained-bundle-v1_0_1.json` and `economics/analytic-correction-v2.json`. Selected SHA-256 identities are:

```
Compiler: capability/compiler.py
7a15a72a727ee9b1c6bb715f0c609bf99194150d156851dcb91eed1f347ebb1c
Scope contract: capability/sensor_profile.json
80aecdfd98dff7dcdc193c591ea6db91ed69edda3571d99d50776f61a521a4df
Archive: capability/retained-v1_0_1.zip
56730d421f19c16044f9e9a57c5d8661c0dfd5f15e8077e3f2ded5c9ff5c74b3
Independent reference: evaluation/reference.py
0f7bce3370e3d0b98450b4a8412c47b37f60b67176c08956737a364454dc386b
Frozen economics: economics/assumptions-v1.json
69f3ad371291069563e751147c219eeea5cb41e0f69a2435830048a4f7d62775
Corrected economics source: economics/economic.py
8a3a83e86f5e14a96f4583c503d2f431a6221980d9e0a12cfe16d1e159a1d412
```

The protected-case manifest hash is recorded with the reference freeze. Protection concerned the development/confirmation sequence; released cases are now public, so rerunning them is reproduction, not another fresh trial. Integration source records were captured after their executions and are explicitly distinguished from the technical prefreeze.

1.3 Result index

- **Origin:** `data/origin.json` and `results/origin.json`. Three operators' expected action expense falls from $1,950 to $526.50 under supplied probabilities, excluding acquisition and fees. Individual information values are $365, $474.50 and $584. The example was deliberately constructed to examine a mechanism, not sampled from a market.
- **Fresh technical work:** `evaluation/confirmation-v1/summary.json` and its per-case receipts. All 48 outputs match the reference: 36 valid missions and 12 refusals. Among valid missions, 20 have positive aggregate information value, four zero and 12 negative. A valid calculation need not justify buying it.
- **Fresh commerce:** `results/economics/confirmation.csv` and `confirmation_summary.json`. The frozen policy places zero acquisition orders across all 48 cases and incurs $346 in modeled preliminary quote costs. `campaign_summary.json` adds the origin contribution without converting unsuccessful assessment into revenue.

- **Transfer:** `evidence/recipient-transfer-v1_0_1/summary.json`. The exact archive runs in a separate isolated-import Python process on unfamiliar case 041. Thirteen controls pass; the independent checker records 174 comparisons and no failures. The ordinary library with identical contents, rights and input produces byte-identical output.
- **Recipient outcome:** Aggregate action expense changes from $236.84 to $100.6016883 before procurement. One operator is worse off even before charges. The recipient declines the commercial offer. Technical reuse does not demonstrate retained profit or consent by every participant.
- **Price diagnostic:** `results/economics/inverse_diagnostic.json`. At the frozen $80 fee, the origin supports acquisition up to $706.3421. Recipient 041 supports at most $0.0418 for its best voluntary subset, still with negative provider contribution; none of the 25 declared price/fee cells is jointly viable. This is a postconfirmation diagnostic, not another protected result.

1.4 Economics and obligations

`economics/assumptions-v1.json` freezes illustrative USD prices: $600 acquisition, $80 per accepted package and $20 customer adaptation. Acceptance is 0.90, common package failure 0.05, precommit withdrawal 0.10 and unpaid-balance default 0.03. These are explicit hypotheses. The fixed fee is neither an optimized tariff nor a discount from a measured human quote.

The primary 44-path event tree is `results/economics/event_tree.json`; `summary.json` is its compact reporting companion. Expected service collections are **$157.2975**, against **$134.4680** in A costs:

Disjoint A component	Expected USD
Preliminary quotes	6.0000
Preparation, construction and correction	43.7400
Per-operator delivery	46.9800
Failed/rejected package handling	3.4061
Accepted-package support	10.0420
Sunk acquisition refunded after common failure	24.3000

Delivery-required learning belongs to A. **B** is additional reusable packaging: $180 initially and $2 per reuse. **C**, if chosen, is $240 of extra research; this is an assumed optional budget, not an observed bill. Neither B nor C is commercially admitted in the primary outcome. At an unmeasured $10 saving per use, 23 compatible uses repay B after upkeep; zero saving rejects investment. Paid provider effort is included in A; unpriced additional obligations and actual research opportunity costs remain unknown. Every additional expected $22.83 of provider cost exhausts the origin's margin.

The accounting separates service fees from acquisition custody and buyer savings. Thirty percent of the service fee is deposited. Common package failure refunds all deposits, leaving the provider with sunk acquisition expense. Individual nonacceptance, including postcommit withdrawal, refunds the fee deposit but retains the acquired-information charge. This is distinct from scientific timeout, which is already in technical expected expense. Default affects the unpaid fee balance. Refund liabilities and unpaid obligations are subtracted before funds become available.

An illustrative all-commit/all-accept/all-pay branch receives $840: $600 pass-through and $240 service revenue. Its A cost is $135 and contribution $105. With seed capital $1,500, terminal cash is $1,605; the $300 reserve leaves $1,305 available. This branch is not the $22.83 expectation. Buyer cash must additionally cover full fees, adaptation and fallback obligations; required balances are $800/$950/$1,100, but buyer financing is unverified. Zero-seed failure is retained in the liquidity analysis. Seed capital, customer savings and tokens are not revenue.

1.5 Preserved failures and corrections

`development/first_counterexample.json` preserves the initial timeout error. Prior risk was 0.00004, below a 0.0001 limit, but risk conditional on timeout was 0.00025778. Using the prior would falsely accept that branch. The corrected compiler conditions on each operator's observation and retains a fixed fallback. This reviewer-suggested counterexample does not measure ordinary model use.

`development/economics-v1/` preserves the initial buyer-liquidity calculation that reserved only the fee deposit. The correction reserves the full fee; a targeted regression refuses buyers whose cash lies between the two requirements. Postcommit withdrawal is also explicitly exercised.

`development/economics-v1_1/` preserves the analytical expertise grid before correcting fee collection, common failure and expected retry cost. The corrected collectible factor is 0.979; joint acceptance equals semantic acceptance after retries times 0.95. An extra $20 per $240 cohort requires 8.9601 percentage points of one-attempt conditional semantic improvement. The correction preserves byte-identical primary summary, event tree, confirmation and campaign outputs; its record includes hand-calculated regressions. It is not a retrospective change to successful primary assumptions.

1.6 Comparators, calibration and scope

H denotes equivalent finite decision-support work, not flight-safety certification. The BLS $134,960 annual wage provides a reference; dividing by assumed 2,080 hours and applying assumed 8–24 hours and 1.5–2.5 burden yields $778.62–$3,893.08. This is an estimated labor scenario, not a market quote. (US Bureau of Labor Statistics 2026)

U, E and E+ are not measured frontier-model trial arms. Their primary acceptance assumptions are equal at 0.90; the sensitivity grid includes equal and worse performance. The competent expert has the same mathematical methods and coordination rights and ties the constructed method before extra B. Component ablations lacking the contingency instrument are not evidence that ordinary users or experts lack it. `evidence/prompt_cards.md` is a prospective protocol authored after confirmation.

Official model documentation establishes identifiers and published prices, not task success or account entitlement. The declared 30,000-input/10,000-output Astra workload costs $0.80 at published rates, already inside A. No direct billed API experiment occurred. `evidence/calibration.md` preserves access limitations, pricing details and source dates. (OpenAI 2026b, 2026c, 2026a)

Conjunction assessment, covariance-aware reasoning and coordinated satellite observations have substantial prior art. NASA guidance, the ESA challenge and federated-network work bound the claim: the contribution is the finite executable witness and accounting/transfer audit, not discovery of Bayesian information value or shared observation. No orbit propagation, fitted sensor likelihoods or external physical validation was executed. (NASA Office of the Chief Engineer 2026; Uriot et al. 2020; European Space Agency 2021)

The retained profile is a **scope contract**, not a learned sensor model. Its bounded limits and Apache-2.0 source rights travel with the two-file archive; protected answers and mission inputs do not. Existing lifecycle fixtures retain identity, revocation, selective rights and separate authority checks. The native scientific scope restriction remains in force. Twenty-nine fixture controls and eight service-ledger reconciliation checks pass; native completeness closes modeled categories, not an actual expense bill. There was no live ENS authentication or production admission. Revocation prevents future mediated retrieval; it cannot erase copies already received.

1.7 Reproduction, budget and release

`evidence/protocol.md` records planned caps of 20 minutes for discovery, 35 implementation, 25 evaluation and 25 writing/demo: 105 minutes total. These are planning caps, not metered effort or a cost receipt. Four materially different opportunities were considered before selection. No paid API or external resource purchase is reported.

The study README and `integration/README.md` specify reproduction and fresh output locations. Preserve archived receipts when rerunning; public cases reproduce published results only. Complete repository qualification is not claimed. The next market-changing test requires a rights-and-timing supplier quote plus two or three operators sharing a permission-cleared retrospective event, with frozen acceptance and fallback terms.

The browser-readable edition and English/French demonstration are local deliverables. Public release requires the repository owner's authorized process and verification of the resulting stable link; no live-publication receipt is supplied here. Earlier freezes and failed trials are preserved. This companion supports review and bounded experimentation, not procurement or spacecraft execution authority.

References

European Space Agency. 2021. "Blockchain Ecosystem for an Autonomous Consensus Mechanism of Federated Satellite Networks." https://ideas.esa.int/core/servlet/hype/IMT?documentId=2d929ce255874896a6a09107ed58060f&documentTableId=7295537402235904412&templateName=&userAction=Browse.

MONTREAL.AI. 2026a. "Invent the Economy: Transformational Intelligence for Creating Viable Markets." https://github.com/MontrealAI/AGI-Alpha-Agent-v0/blob/231834269599d85e49cfe1ca9917a1a874522d44/docs/Invent_the_Economy.tex.

———. 2026b. "Transformational Autonomous Enterprises: SUCCESSOR Omega Harness." https://github.com/MontrealAI/AGI-Alpha-Agent-v0/blob/231834269599d85e49cfe1ca9917a1a874522d44/docs/papers/transformational-autonomous-enterprises-v1.tex.

———. 2026c. "Transformational Intelligence: Learning to Change the Game." https://github.com/MontrealAI/AGI-Alpha-Agent-v0/blob/231834269599d85e49cfe1ca9917a1a874522d44/docs/Transformational_Intelligence.tex.

NASA Office of the Chief Engineer. 2026. "Spacecraft Conjunction Assessment and Collision Avoidance Best Practices Handbook: Volume 1." https://nodis3.gsfc.nasa.gov/OCE_docs/OCE_51.pdf.

OpenAI. 2026a. "API Pricing." https://developers.openai.com/api/docs/pricing.

———. 2026b. "GPT-6 Astra Model." https://developers.openai.com/api/docs/models/gpt-6-astra.

———. 2026c. "GPT-6.1 Terra Model." https://developers.openai.com/api/docs/models/gpt-6.1-terra.

Uriot, Thomas, Dario Izzo, Luís F. Simões, Rasit Abay, Nils Einecke, Sven Rebhan, Jose Martinez-Heras, Francesca Letizia, Jan Siminski, and Klaus Merz. 2020. "Spacecraft Collision Avoidance Challenge: Design and Results of a Machine Learning Competition." *arXiv:2008.03069*. https://arxiv.org/abs/2008.03069.

US Bureau of Labor Statistics. 2026. "Aerospace Engineers: Occupational Outlook Handbook." https://www.bls.gov/ooh/architecture-and-engineering/aerospace-engineers.htm.